

Why Do Lottery Tickets Need Learning-Rate Warmup?

Sharpness, weight correlation, or the mask: a causal test on ResNet-20 / CIFAR-10

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Abstract

Frankle & Carbin (2019) found that deep networks only yield winning lottery tickets at a small learning rate or with learning-rate warmup, and left the reason open. We test two explanations from the literature. The first says that without warmup, early training crosses the stability limit $\eta \cdot \lambda_{\max} \approx 2 + 2\beta$ (sharpness). The second, from Liu et al. (2021), says that a large learning rate decorrelates the trained weights from their initialization (weight correlation). On ResNet-20 / CIFAR-10 with iterative magnitude pruning down to 6% of weights, we replicate the phenomenon on two seeds. Tickets beat trained baselines by +2 to +6.4 pp at $\eta = 0.01$ and at $\eta = 0.03$ with warmup, but by at most +0.5 pp at $\eta = 0.1$ without warmup. Neither explanation survives. No run comes near the stability limit ($\max S = 0.54$). In a pre-registered causal test, the good masks found with warmup, trained at $\eta = 0.1$ with no warmup, **still win** (+3.6 and +4.8 pp over shuffled controls), despite the same early sharpness transient and near-chance weight correlation. Lowering sharpness with SAM changes nothing, and forcing correlation with an L2 anchor does not improve the ticket. The result replicates on a second seed, and the reverse test confirms it: the masks found without warmup still fail when trained *with* warmup. High-learning-rate training does not break the ticket. It breaks **iterative magnitude pruning’s choice of mask**.

1. Introduction

The lottery ticket hypothesis (Frankle & Carbin, 2019) states that a dense network contains a sparse subnetwork, the *winning ticket*. Reset to its original initialization θ_0 and trained alone, the ticket matches the dense network. Tickets are found by iterative magnitude pruning (IMP): train, prune the smallest weights, rewind the survivors to θ_0 , repeat. On deeper networks (VGG-19, ResNet-18) this only works at a small learning rate or with warmup; at the standard rate the “ticket” is no better than a random reinitialization. Why warmup is needed was left open.

Two later lines of work give candidate answers:

- **Sharpness / stability.** Gradient descent is stable only while $\eta \cdot \lambda_{\max}$ stays below a threshold ($2 + 2\beta$ for heavy-ball momentum). Cohen et al. (2021) showed that training self-stabilizes at this “edge of stability”, and Kalra & Barkeshli (2024) showed that warmup acts by letting the network move to flatter regions before the learning rate is large. If a sparse ticket starts sharper, or crosses the threshold early, its training might be disrupted in a way that loses what made it special.
- **Weight correlation.** Liu et al. (2021) argue that tickets exist only when training keeps the final weights θ_T correlated with θ_0 (overlap R_p of the largest weights). A large learning rate moves the weights further and destroys that correlation.

The observational data in this setting cannot separate these explanations: in Frankle’s results, warmup lowers early sharpness **and** preserves correlation **and** lowers the average learning rate. We therefore add a causal test in which the mask is held fixed and each mechanism is manipulated on its own.

2. Hypotheses

- **H1 (replication).** On ResNet-20 / CIFAR-10, tickets beat a randomly reinitialized network at $\eta = 0.01$ and with warmup ($\eta = 0.03$), but not at $\eta = 0.1$ without warmup.
- **H2 (stability ratio).** Where tickets fail, the stability ratio $S(t) = \eta_t \cdot \lambda_{\max}(t) / (2 + 2\beta)$, measured in early training, reaches about 1. Where tickets win, S stays clearly below 1. Sharpness at initialization alone is not expected to predict this.
- **H3 (warmup mechanism).** Warmup keeps S below 1 through the high-learning-rate phase, and this coincides with the return of the ticket advantage.
- **H4 (causal test).** Take the good masks from the warmup chain and train them at $\eta = 0.1$ (a) with no warmup, (b) with SAM, which lowers sharpness, and (c) with an L2 anchor to θ_0 , which raises correlation. If (b) rescues the ticket and (c) does not, that supports sharpness; the reverse supports Liu et al.

3. Methods

Data and model. CIFAR-10 is split into 45,000 training and 5,000 validation images, plus the official 10,000-image test set (used only for reported numbers). Augmentation is random crop with 4-px padding and horizontal flip. The model is ResNet-20 (3 stages $\times$ 3 basic blocks, 16/32/64 channels, BatchNorm, 0.27 M parameters).

Training. SGD with momentum $\beta = 0.9$, batch 128, weight decay 10^{-4} , 15,000 iterations ($\approx$ 38 epochs), learning rate $\times 0.1$ at 10k and 12.5k, fp16 mixed precision. Four conditions:

Condition	η	Warmup	Role
low	0.01	none	tickets expected to win
high	0.1	none	tickets expected to fail
warm03	0.03	linear over 10k steps	Frankle’s successful ResNet setting
high_warm	0.1	linear over 10k steps	the same peak rate as high, with warmup

IMP. Each round prunes 30% of the surviving conv and linear weights globally by magnitude; the final layer is pruned at half the rate and the first conv layer is never pruned. The survivors are reset to the exact θ_0 (no re-winding). Eight rounds give 100% down to 6.0% of weights remaining. The mask is re-applied to the weights and the momentum buffers after every step.

Baselines. At the same mask: **reinit** (fresh random initialization; rounds 2, 4, 6, 8) and **shuffle** (θ_0 permuted within each layer over the surviving positions, which keeps each layer’s weight distribution; rounds 4 and 8). The **ticket advantage** is the ticket’s test accuracy minus the best trained baseline’s. A **winning ticket** needs advantage > 0 (with two seeds, mean > 2 SE) *and* accuracy within 0.5 pp of the dense network of its condition.

Sharpness. $\lambda_{\max}$ of the training loss on a fixed batch of 2,048 images, by 10-step Lanczos with full reorthogonalization on masked Hessian-vector products (fp32). It is measured at steps 0, 25, 50, 100, 200, every 500

to 3,000, then every 3,000 (15 points per run). The H2 statistic is **max S over steps 25–3,000**; $S(0)$ is reported separately because it measures the initialization, not training. (Power iteration converged to the most negative eigenvalue on ResNet-20, so we switched to Lanczos.)

Weight correlation. R_p (Liu et al., 2021): the share of the top- p surviving weights by $|\theta_0|$ that are also in the top- p by $|\theta_T|$, pooled over layers. Chance is $\approx p$; we report $R_{0.2}$.

H4 design. Masks and θ_0 come from warm03 seed 0 at rounds 3, 6 and 8 (34.5, 12.0 and 6.0% remaining). Each mask is trained as ticket and shuffle at $\eta = 0.1$ with no warmup: (a) plain, (b) SAM with $p = 0.05$, the perturbation restricted to the mask, and (c) an L2 anchor $(\lambda/2) \cdot \|m \odot (\theta - \theta_0)\|^2$. λ was chosen before the main runs as the smallest value in $\{10^{-3}, 10^{-2}\}$ whose $R_{0.2}$ at step 3,000 reached the low condition’s level (0.661); $\lambda = 10^{-2}$ reached 0.757. **The decision rule was fixed and committed before any H4 result:** an intervention rescues the ticket if ticket – shuffle $\geq +2.0$ pp at ≥ 2 of the 3 masks, *and* it moved its target variable relative to (a) (SAM lowers max S; the anchor raises $R_{0.2}$ at step 3,000). The 2.0 pp bar sits above everything the failing high chain produced ($\leq +0.7$ pp).

Scale and compute. Seeds 0 and 1 for low, high and warm03; seed 0 for high_warm and H4, plus two pre-registered follow-ups (H4(a) on seed 1, and a reverse swap; §4.5); 137 trainings in all (including 2 short runs that chose the anchor strength), on one NVIDIA T4 (Colab), three processes in parallel under CUDA MPS. Total ≈ 26 hours of a 30-hour runtime budget.

4. Results

4.1 H1: replication, supported

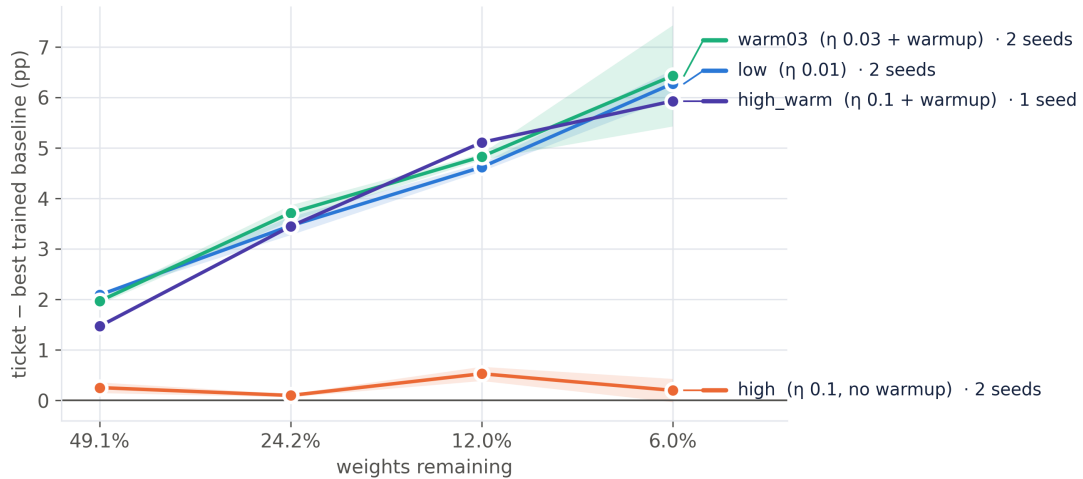


Figure 1. Ticket advantage over the best trained baseline. Lines are two-seed means (one seed for high_warm), bands ± 1 SE.

At $\eta = 0.01$ and at $\eta = 0.03$ with warmup, tickets beat their trained baselines at every sparsity, by more as sparsity grows: low +2.09 / +3.46 / +4.62 / +6.27 pp and warm03 +1.97 / +3.72 / +4.83 / +6.43 pp at 49.1 / 24.2 / 12.0 / 6.0% remaining (two-seed means). At $\eta = 0.1$ without warmup the advantage collapses to +0.26 / +0.10 / +0.53 / +0.20 pp, with tickets 0.8 to 7.2 pp below their dense network: **no winning ticket on either seed**. Adding warmup at the same peak rate (high_warm) restores it: +1.47 / +3.45 / +5.11 / +5.93 pp, winning at 3 of 4 sparsities. The two seeds agree within 0.6 pp on every advantage except warm03 at 6.0% (+7.43 vs +5.43 pp).

Table 1 gives the rubric’s metrics for the dense networks and two ticket sparsities (test set, mean $\pm$ SD over seeds).

Condition	Network	Acc	Precision	Recall	F1	AUC
low	dense	86.57 $\pm$ 0.13	86.59	86.57	86.56	0.9894
low	ticket 34.5%	88.00 $\pm$ 0.18	87.99	88.00	87.98	0.9915
low	ticket 6.0%	86.18 $\pm$ 0.06	86.15	86.18	86.12	0.9888
high	dense	89.69 $\pm$ 0.20	89.66	89.69	89.67	0.9933
high	ticket 34.5%	88.23 $\pm$ 0.18	88.21	88.23	88.20	0.9918
high	ticket 6.0%	82.54 $\pm$ 0.25	82.48	82.54	82.46	0.9830
warm03	dense	87.64 $\pm$ 0.23	87.62	87.64	87.62	0.9907
warm03	ticket 34.5%	88.79 $\pm$ 0.37	88.77	88.79	88.76	0.9925
warm03	ticket 6.0%	87.43 $\pm$ 0.60	87.40	87.43	87.38	0.9907
high_warm	dense	89.46	89.49	89.46	89.46	0.9932
high_warm	ticket 34.5%	90.30	90.30	90.30	90.29	0.9943
high_warm	ticket 6.0%	88.35	88.39	88.35	88.35	0.9924

Table 1. Accuracy (mean $\pm$ SD over seeds), macro precision / recall / F1 (%) and macro one-vs-rest ROC-AUC. Two seeds except high_warm.

The highest-accuracy networks are trained at $\eta = 0.1$ (dense high 89.7%, high_warm tickets up to 90.3%), so the failure at $\eta = 0.1$ is not a weak dense baseline: the dense network does well, and only IMP’s tickets lose their edge.

4.2 H2: stability ratio, not supported

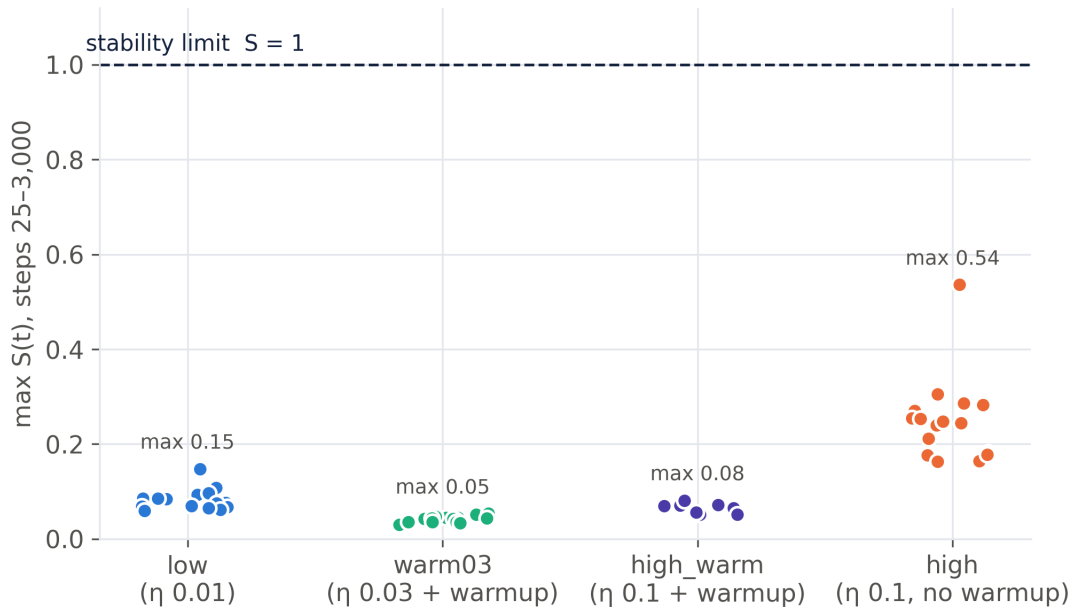


Figure 2. Max $S(t)$ over steps 25–3,000 for every ticket, by condition. The dashed line is the stability limit $S = 1$.

- **No run reaches the limit.** Over 63 tickets and 42 baselines, the largest max $S(25-3k)$ is 0.54. The failing high tickets peak at 0.16–0.54, a factor of 2 to 6 below $S = 1$.
- **The failing condition does stand apart.** The high tickets (0.16–0.54) do not overlap any winning condition (≤ 0.15), and across all tickets the Spearman correlation between max S and the ticket’s gap to dense is ρ

= -0.59 (n = 56). The difference is a brief transient: in 15 of 18 high tickets the maximum falls at step 25, and after step 1,000 all conditions sit at $S \approx 0.02$ –0.09.

- **Within a condition, S does not order the tickets** ($\rho = +0.26$ for high, -0.38 for low, $+0.19$ for warm03). In high the accuracy deficit grows from -0.3 to -7.1 pp with sparsity while max S stays flat. $S(0)$ even runs the wrong way (sparser masks start flatter and fail worse).
- **Caveat on the threshold.** With momentum, the step-0 threshold is $2/\lambda$, not $(2 + 2\beta)/\lambda$, because the momentum buffer starts empty (Kalra & Barkeshli, 2024). Against $2/\lambda$, the dense high network does start above the limit (1.27 on seed 0, 2.03 on seed 1) and trains normally anyway. The minibatch threshold later in training is lower than the full-batch one and was not measured.

So H2 fails as stated. A weaker, condition-level association holds, and H4 tests whether it is causal.

4.3 H3: warmup mechanism, partly supported

At the same peak rate $\eta = 0.1$, warmup brings the ticket advantage back (high_warm wins 3 of 4, high 0 of 4) and removes the step-25 transient (S at step $25 \leq 0.003$ with warmup vs 0.13 – 0.54 without). But “keeps S below 1” does not discriminate, because high also stays below 1, and during the high-learning-rate phase itself the two runs look alike (at step 9,000: $\lambda_{\max}$ 1.1–1.7, S 0.03–0.04 in both). Warmup also lets sharpness first grow (warm03: $\lambda_{\max}$ 25.5 $\rightarrow$ 52.7 by step 2,000) before it falls, which is the progressive sharpening Kalra & Barkeshli describe. The Phase A data cannot separate the transient from the other things warmup changes: it also keeps more weight correlation (dense $R_{0.2}$ 0.33 vs 0.25) and halves the mean learning rate over the first 10k steps.

4.4 H4: the causal test

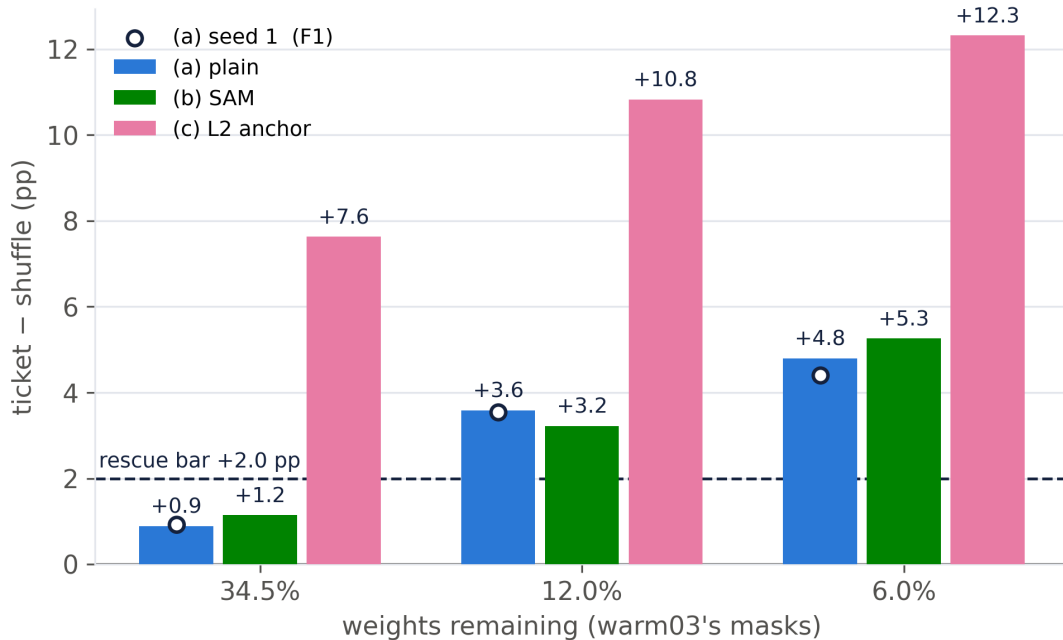


Figure 3. Ticket – shuffle advantage on warm03's three masks, trained at $\eta = 0.1$ without warmup. Dashed line: the pre-registered +2.0 pp rescue bar. Open circles: (a) repeated on seed 1 (F1, \$4.5).

Mask	(a) plain	(b) SAM	(c) L2 anchor
34.5%	89.27 / 88.38 / +0.89	89.29 / 88.14 / +1.15	85.64 / 78.01 / +7.63
12.0%	88.91 / 85.32 / +3.59	88.86 / 85.64 / +3.22	86.19 / 75.36 / +10.83
6.0%	87.24 / 82.44 / +4.80	87.50 / 82.24 / +5.26	84.44 / 72.12 / +12.32
max S(25–3k), tickets	0.34 / 0.27 / 0.24	0.22 / 0.21 / 0.19	0.31 / 0.25 / 0.24
$R_{0.2}$ at step 3k, tickets	0.30 / 0.22 / 0.22	0.30 / 0.21 / 0.21	0.75 / 0.63 / 0.54

Table 2. Ticket / shuffle test accuracy (%) and advantage (pp) at each mask (seed 0), with the manipulation-check variables.

(a) The good masks still win at $\eta = 0.1$ without warmup: +3.59 and +4.80 pp at 12.0 and 6.0%, 2 of 3 masks over the bar. The pre-registered reading of this outcome is that the failure of the high chain does not come from how a good mask trains at $\eta = 0.1$. It comes from the masks IMP finds at $\eta = 0.1$. Figure 4 shows this directly. Under identical training ($\eta = 0.1$, no warmup, same θ_0), warm03's 6.0% mask reaches 87.24% and high's own 6.0% mask 82.72%, while the shuffled controls on the two masks are equal (82.4% vs 82.2%). Both readings from Section 1 fail here:

- **the early sharpness transient is present** (max S 0.24–0.34, inside the failing high chain's range and 6–9× warm03's), and the ticket still wins;
- **the weight correlation is lost** (final $R_{0.2}$ 0.18–0.20, chance level, no higher than in the failing high chain), and the ticket still wins.

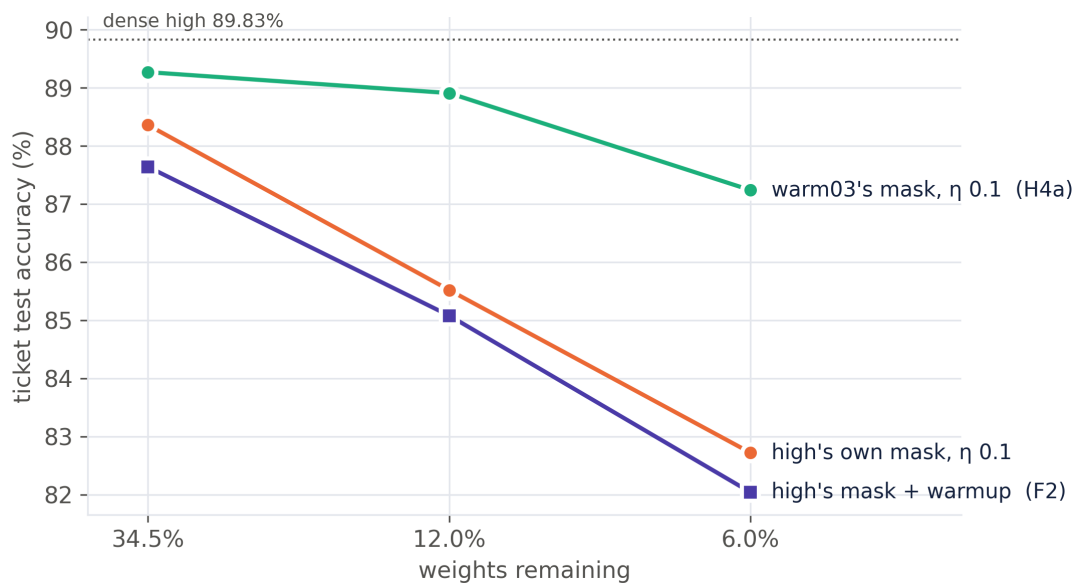


Figure 4. Ticket accuracy on three masks under the same training. The green and orange lines differ only in the mask (same θ_0 , $\eta = 0.1$, no warmup); the orange and violet lines differ only in warmup (F2, §4.5). The mask decides.

(b) SAM lowered max S below (a) at all three masks (the manipulation worked) and left the advantage where it was ($\Delta \leq 0.5$ pp; ticket accuracy within 0.3 pp of (a)). Lowering sharpness after the first steps changes neither the ticket nor its advantage.

(c) The L2 anchor raised $R_{0.2}$ well above (a) at all three masks (the manipulation worked) and produced the largest advantages, +7.6 to +12.3 pp. But the ticket got worse: the anchored tickets lose 2.7–3.6 pp relative to (a), and the anchored shuffles lose 10.0–10.4 pp. Pulling the weights toward a shuffled θ_0 hurts far more than pulling them toward the ticket's own θ_0 . That confirms that the ticket's θ_0 suits its mask, which is the lottery ticket hypothesis itself, but forcing correlation does not make a better ticket.

By the rule, (b) and (c) both formally “rescue”. Since (a) already meets the rule, there is no failure for them to rescue, and the pre-registered analysis treats them as secondary.

4.5 Follow-up runs

The H4 verdict rested on one seed, and “the mask is the cause” was inferred rather than tested. We pre-registered and ran two follow-ups (same code, same +2.0 pp rule) before seeing their results.

Run	34.5%	12.0%	6.0%	Masks $\geq$ +2.0 pp
(a) seed 0 · warm03 masks, η 0.1	+0.89	+3.59	+4.80	2 / 3
F1 · the same on seed 1	+0.93	+3.54	+4.40	2 / 3
high chain · its own masks, η 0.1	—	+0.39	+0.57	0 / 2
F2 · high’s masks, η 0.1 + warmup	−0.02	+0.67	+0.31	0 / 3

Table 3. Ticket – baseline advantage (pp) in the follow-up runs. F1 uses warm03 seed 1’s masks and θ_0 ; F2 uses high’s seed-0 masks and the high_warm schedule. The high-chain row is from Phase A (reinit at 12.0%, shuffle at 6.0%).

F1, replication: confirmed. On an independent seed, warm03’s masks win at $\eta = 0.1$ without warmup at 2 of 3 masks, with advantages within 0.4 pp of seed 0 at every mask. As in seed 0, the early sharpness transient is present (max S 0.25–0.27) and $R_{0.2}$ is at chance (0.18–0.20).

F2, reverse swap: warmup does not rescue a bad mask. high’s own masks, trained from the same θ_0 on the high_warm schedule, stay at or below +0.7 pp, which is the failing chain’s level. Warmup did what it does elsewhere: it removed the early transient (max S 0.09–0.12, against 0.18–0.29 without warmup) and kept slightly more correlation ($R_{0.2}$ 0.25–0.28 vs 0.22–0.23). Yet the tickets were no better (87.64 / 85.08 / 82.04% vs 88.36 / 85.52 / 82.72% without warmup).

Together, with θ_0 and $\eta = 0.1$ held fixed, **a good mask wins without warmup, and a bad mask loses with it.** Warmup during training is neither necessary for a good mask nor sufficient for a bad one.

5. Discussion

What the evidence supports. Tickets fail at $\eta = 0.1$ without warmup (H1), but the stability ratio never reaches 1 (H2), and warmup’s role is not to train a given mask more stably (H3). The causal test points away from training dynamics altogether. A good mask trains to a winning ticket at $\eta = 0.1$, through the same early sharpness spike and the same loss of correlation that accompany the failing chain. What differs is the mask. The high learning rate damages **IMP’s selection of the mask**: the magnitudes after a high-learning-rate training run are a poor guide to which weights should survive, and pruning on them compounds over rounds. This fits Paul et al. (2023), who show that an IMP mask encodes information about the training run that produced it, and it reframes Frankle & Carbin’s warmup observation: warmup matters while the mask is being *found*. The reverse swap (F2) tests this directly: applying warmup only when the bad mask is trained, after the mask has been found, does not help.

Why the two literature explanations looked plausible. In the observational data every variable moves together: the failing condition has the early sharpness spike, the lowest correlation, and the bad masks. Only by holding the mask fixed and varying the training does the confound break. This is the main methodological lesson of the project.

Limitations.

- **Seeds.** Two seeds for the main conditions; one for high_warm, SAM, the anchor and the reverse swap. The central H4(a) result holds on two seeds (F1).
- **Strict definition.** Under the winning-ticket definition (within 0.5 pp of dense), the plain $\eta = 0.1$ tickets on warm03's masks are still 0.56, 0.92 and 2.59 pp below dense high (89.83%). Some cost of $\eta = 0.1$ survives a good mask, and the advantage is smaller than in warm03 itself (+3.59 vs +4.89 pp at 12.0%).
- **Shortened protocol.** 15k iterations (Frankle uses 30k), 30% pruning per round (Frankle 20%), no rewinding, one architecture and one dataset.
- **Sharpness sampling.** 15 points per run on a fixed 2,048-image batch; transients between points (especially within the first 25 steps) are missed, and the minibatch stability threshold was not measured.
- **Interventions at one strength.** SAM at $\rho = 0.05$ and the anchor at $\lambda = 10^{-2}$ only.

Next steps. (1) Test which *round* of IMP at $\eta = 0.1$ first produces a bad mask, by switching the learning rate mid-chain. (2) Compare high and warm03 masks directly (layer-wise density, overlap) to find what distinguishes them. (3) Repeat with learning-rate rewinding (Frankle et al., 2020), which is known to restore tickets at high rates and would test the mask-selection reading.

6. Conclusion

We replicated the lottery ticket learning-rate effect on ResNet-20 / CIFAR-10 on two seeds and tested two mechanistic explanations with a pre-registered causal experiment. Neither sharpness nor weight correlation explains why tickets fail at a high learning rate without warmup: a good mask still wins under exactly those training conditions (on two seeds), a bad mask still fails when warmup is added, and directly lowering sharpness or raising correlation does not improve the ticket. The failure lies in the mask that iterative magnitude pruning selects when the network is trained at a high learning rate.

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Appendix: reproducibility

All code, configs and per-run results are in the repository. `python -m analysis.plots` regenerates every table and figure from `results/` (`summary.csv`, `ticket_advantage.csv`, `gate_a.md`, `h4_results.md`, figures 1-5); `python -m analysis.final_figs` regenerates the figures in this report. Decisions and deviations from the original plan (Lanczos instead of power iteration, the sharpness schedule, the H2 statistic excluding step 0, the Gate A and H4 decision rules) are listed in `Methodology.md` §9; the H4 pre-registration is in `results/H4.md` §1-5 and §8, committed before the corresponding runs.